Starhaven Master System Specification

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Author	Idris Kalvane
Approved by	Edmund Tarrow
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1. Introduction

This specification is the top-level requirements document for Starhaven Orbital Habitat & Shipyard. Each chapter covers one station system; each section covers one component, its requirements, its interfaces and its zone allocation.

System	Name	Lead department	Components
HR	Hypermatter Reactor	Reactor & Power Systems	5
TX	Thermal Control & Heat Rejection	Reactor & Power Systems	4
PD	Power Distribution	Reactor & Power Systems	4
SA	Solar Collector Array	Reactor & Power Systems	3
PR	Propulsion & Station-Keeping	Propulsion & Station-Keeping	4
HB	Docking & Hangar Bays	Docking & Flight Operations	4
SY	Shipyard Slipways	Shipyard Operations	5
LS	Life Support & Atmospherics	Life Support & Habitation	4
HA	Habitation Blocks	Life Support & Habitation	5
WM	Waste Management & Recycling	Life Support & Habitation	3
AG	Hydroponics & Food Production	Agriculture & Food Systems	4
CN	Computer Core & Data Network	Computing & Communications	4
CM	Communications Relay	Computing & Communications	3
SF	Superstructure & Frame	Structural Engineering	4
TR	Internal Transit	Structural Engineering	4
RL	Research Laboratories	Science & Research	4
OC	Operations Centre	Program Management Office	3

2. Hypermatter Reactor (HR)

The hypermatter reactor is the station's main power plant. It supplies every zone substation through the power distribution system and is backed by the solar collector array.

Document	Title
SHV-ICD-HR-TX	Interface Control Document: Hypermatter Reactor / Thermal Control & Heat Rejection
SHV-ICD-HR-PD	Interface Control Document: Hypermatter Reactor / Power Distribution
SHV-ICD-HR-CN	Interface Control Document: Hypermatter Reactor / Computer Core & Data Network

2.1 Reactor core vessel (HR-RCV)

Detailed requirements: SHV-HR-SPEC-0002 Rev D. Drawing: SHV-HR-DWG-0201. Bill of materials: SHV-HR-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
HR-RCV-R01	Core diameter	1.85 km	±0.5%	Inspection
HR-RCV-R02	Wall thickness	18.7 m	±0.5%	Demonstration
HR-RCV-R03	Design core temperature	4,400 K	±0.5%	Test

Verification approach

- HR-RCV-R01: inspection per the station verification and test plan.
- HR-RCV-R02: demonstration per the station verification and test plan.
- HR-RCV-R03: test per the station verification and test plan.

2.1.1 Zone allocation and installation

The reactor core vessel is installed in zones Z05, Z24.

Zone	Sequence	Installed by	Inspection
Z05	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z24	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

2.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

2.1.3 Applicable documents

Document	Revision	Title
SHV-HR-SPEC-0002	Rev D	Component specification
SHV-HR-DWG-0201	current	General arrangement drawing
SHV-HR-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

2.1.4 General requirements

ID	Requirement
HR-RCV-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HR-RCV-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HR-RCV-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HR-RCV-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HR-RCV-G05	Documentation: Acceptance data packages shall be delivered with every unit.

2.2 Containment field generator (HR-CFG)

Detailed requirements: SHV-HR-SPEC-0102 Rev C. Drawing: SHV-HR-DWG-0202. Bill of materials: SHV-HR-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
HR-CFG-R01	Field strength	96.66 % nominal	±5%	Test
HR-CFG-R02	Generator count	19 ea	±2.5%	Inspection

Verification approach

- HR-CFG-R01: test per the station verification and test plan.
- HR-CFG-R02: inspection per the station verification and test plan.

2.2.1 Zone allocation and installation

The containment field generator is installed in zones Z02, Z08, Z22, Z24.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z22	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z24	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

2.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

2.2.3 Applicable documents

Document	Revision	Title
SHV-HR-SPEC-0102	Rev C	Component specification
SHV-HR-DWG-0202	current	General arrangement drawing
SHV-HR-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

2.2.4 General requirements

ID	Requirement
HR-CFG-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HR-CFG-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HR-CFG-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HR-CFG-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HR-CFG-G05	Documentation: Acceptance data packages shall be delivered with every unit.

2.3 Fuel injector assembly (HR-FIA)

Detailed requirements: SHV-HR-SPEC-0103 Rev A. Drawing: SHV-HR-DWG-0203. Bill of materials: SHV-HR-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
HR-FIA-R01	Injection rate	7.79 kg/s	±2.5%	Demonstration
HR-FIA-R02	Injector count	128 ea	±1%	Analysis

Verification approach

- HR-FIA-R01: demonstration per the station verification and test plan.
- HR-FIA-R02: analysis per the station verification and test plan.

2.3.1 Zone allocation and installation

The fuel injector assembly is installed in zones Z10, Z13, Z22.

Zone	Sequence	Installed by	Inspection
Z10	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z13	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z22	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

2.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

2.3.3 Applicable documents

Document	Revision	Title
SHV-HR-SPEC-0103	Rev A	Component specification
SHV-HR-DWG-0203	current	General arrangement drawing
SHV-HR-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

2.3.4 General requirements

ID	Requirement
HR-FIA-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HR-FIA-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HR-FIA-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HR-FIA-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HR-FIA-G05	Documentation: Acceptance data packages shall be delivered with every unit.

2.4 Coolant manifold (HR-CMF)

Detailed requirements: SHV-HR-SPEC-0104 Rev C. Drawing: SHV-HR-DWG-0204. Bill of materials: SHV-HR-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
HR-CMF-R01	Flow capacity	964 m³/s	±1%	Test
HR-CMF-R02	Pressure rating	42 MPa	±5%	Analysis

Verification approach

- HR-CMF-R01: test per the station verification and test plan.
- HR-CMF-R02: analysis per the station verification and test plan.

2.4.1 Zone allocation and installation

The coolant manifold is installed in zones Z01, Z19, Z20, Z23.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z19	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z20	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z23	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

2.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

2.4.3 Applicable documents

Document	Revision	Title
SHV-HR-SPEC-0104	Rev C	Component specification
SHV-HR-DWG-0204	current	General arrangement drawing
SHV-HR-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

2.4.4 General requirements

ID	Requirement
HR-CMF-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HR-CMF-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HR-CMF-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HR-CMF-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HR-CMF-G05	Documentation: Acceptance data packages shall be delivered with every unit.

2.5 Reactor monitoring station (HR-RMS)

Detailed requirements: SHV-HR-SPEC-0105 Rev B. Drawing: SHV-HR-DWG-0205. Bill of materials: SHV-HR-BOM-0305.

Requirement	Parameter	Value	Tolerance	Verification
HR-RMS-R01	Sensor channels	5,286 ea	±0.5%	Demonstration
HR-RMS-R02	Scan interval	45 ms	±0.5%	Demonstration

Verification approach

- HR-RMS-R01: demonstration per the station verification and test plan.
- HR-RMS-R02: demonstration per the station verification and test plan.

2.5.1 Zone allocation and installation

The reactor monitoring station is installed in zones Z07, Z21.

Zone	Sequence	Installed by	Inspection
Z07	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z21	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

2.5.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

2.5.3 Applicable documents

Document	Revision	Title
SHV-HR-SPEC-0105	Rev B	Component specification
SHV-HR-DWG-0205	current	General arrangement drawing
SHV-HR-BOM-0305	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

2.5.4 General requirements

ID	Requirement
HR-RMS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HR-RMS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HR-RMS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HR-RMS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HR-RMS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

3. Thermal Control & Heat Rejection (TX)

Thermal control carries surplus reactor and habitat heat to the radiator fin arrays on the station surface through heat exchanger stacks, pumped coolant loops and passive heat pipes.

Document	Title
SHV-ICD-HR-TX	Interface Control Document: Hypermatter Reactor / Thermal Control & Heat Rejection

3.1 Heat exchanger stack (TX-HXS)

Detailed requirements: SHV-TX-SPEC-0101 Rev C. Drawing: SHV-TX-DWG-0201. Bill of materials: SHV-TX-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
TX-HXS-R01	Rejection capacity	0.66 PW	±5%	Test
TX-HXS-R02	Stack count	20 ea	±2%	Analysis

Verification approach

- TX-HXS-R01: test per the station verification and test plan.
- TX-HXS-R02: analysis per the station verification and test plan.

3.1.1 Zone allocation and installation

The heat exchanger stack is installed in zones Z21, Z23.

Zone	Sequence	Installed by	Inspection
Z21	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z23	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

3.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

3.1.3 Applicable documents

Document	Revision	Title
SHV-TX-SPEC-0101	Rev C	Component specification
SHV-TX-DWG-0201	current	General arrangement drawing
SHV-TX-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

3.1.4 General requirements

ID	Requirement
TX-HXS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TX-HXS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TX-HXS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TX-HXS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TX-HXS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

3.2 Radiator fin array (TX-RFA)

Detailed requirements: SHV-TX-SPEC-0007 Rev C. Drawing: SHV-TX-DWG-0202. Bill of materials: SHV-TX-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
TX-RFA-R01	Fin area	6.3 km²	±2%	Demonstration
TX-RFA-R02	Emissivity	0.89	±5%	Demonstration

Verification approach

- TX-RFA-R01: demonstration per the station verification and test plan.
- TX-RFA-R02: demonstration per the station verification and test plan.

3.2.1 Zone allocation and installation

The radiator fin array is installed in zones Z02, Z17.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z17	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

3.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

3.2.3 Applicable documents

Document	Revision	Title
SHV-TX-SPEC-0007	Rev C	Component specification
SHV-TX-DWG-0202	current	General arrangement drawing
SHV-TX-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

3.2.4 General requirements

ID	Requirement
TX-RFA-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TX-RFA-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TX-RFA-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TX-RFA-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TX-RFA-G05	Documentation: Acceptance data packages shall be delivered with every unit.

3.3 Coolant pump station (TX-CPS)

Detailed requirements: SHV-TX-SPEC-0103 Rev C. Drawing: SHV-TX-DWG-0203. Bill of materials: SHV-TX-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
TX-CPS-R01	Pump flow	104 m³/s	±2%	Test
TX-CPS-R02	Pump count	12 ea	±2.5%	Inspection

Verification approach

- TX-CPS-R01: test per the station verification and test plan.
- TX-CPS-R02: inspection per the station verification and test plan.

3.3.1 Zone allocation and installation

The coolant pump station is installed in zones Z02, Z07, Z16, Z17, Z23.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z07	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z16	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z17	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z23	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

3.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

3.3.3 Applicable documents

Document	Revision	Title
SHV-TX-SPEC-0103	Rev C	Component specification
SHV-TX-DWG-0203	current	General arrangement drawing
SHV-TX-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

3.3.4 General requirements

ID	Requirement
TX-CPS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TX-CPS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TX-CPS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TX-CPS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TX-CPS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

3.4 Heat pipe loop (TX-HPL)

Detailed requirements: SHV-TX-SPEC-0104 Rev A. Drawing: SHV-TX-DWG-0204. Bill of materials: SHV-TX-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
TX-HPL-R01	Loop length	5.78 km	±2.5%	Analysis
TX-HPL-R02	Operating temperature	282 K	±5%	Test

Verification approach

- TX-HPL-R01: analysis per the station verification and test plan.
- TX-HPL-R02: test per the station verification and test plan.

3.4.1 Zone allocation and installation

The heat pipe loop is installed in zones Z06, Z12, Z14, Z20.

Zone	Sequence	Installed by	Inspection
Z06	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z12	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z14	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z20	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

3.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

3.4.3 Applicable documents

Document	Revision	Title
SHV-TX-SPEC-0104	Rev A	Component specification
SHV-TX-DWG-0204	current	General arrangement drawing
SHV-TX-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

3.4.4 General requirements

ID	Requirement
TX-HPL-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TX-HPL-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TX-HPL-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TX-HPL-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TX-HPL-G05	Documentation: Acceptance data packages shall be delivered with every unit.

4. Power Distribution (PD)

Power distribution carries reactor and solar output to the zone substations over superconducting bus conduits, with backup generator banks and storage battery strings for life-critical loads.

Document	Title
SHV-ICD-HR-PD	Interface Control Document: Hypermatter Reactor / Power Distribution
SHV-ICD-SA-PD	Interface Control Document: Solar Collector Array / Power Distribution
SHV-ICD-PD-LS	Interface Control Document: Power Distribution / Life Support & Atmospherics
SHV-ICD-PD-HB	Interface Control Document: Power Distribution / Docking & Hangar Bays
SHV-ICD-PD-SY	Interface Control Document: Power Distribution / Shipyard Slipways
SHV-ICD-PD-PR	Interface Control Document: Power Distribution / Propulsion & Station-Keeping
SHV-ICD-PD-AG	Interface Control Document: Power Distribution / Hydroponics & Food Production
SHV-ICD-PD-TR	Interface Control Document: Power Distribution / Internal Transit
SHV-ICD-PD-RL	Interface Control Document: Power Distribution / Research Laboratories

4.1 Main bus conduit (PD-MBC)

Detailed requirements: SHV-PD-SPEC-0101 Rev A. Drawing: SHV-PD-DWG-0201. Bill of materials: SHV-PD-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
PD-MBC-R01	Conduit rating	10.3 % reactor	±0.5%	Inspection
PD-MBC-R02	Conduit length	57 km	±2.5%	Test

Verification approach

- PD-MBC-R01: inspection per the station verification and test plan.
- PD-MBC-R02: test per the station verification and test plan.

4.1.1 Zone allocation and installation

The main bus conduit is installed in zones Z02, Z11, Z13, Z16, Z20.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z11	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z13	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z16	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z20	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

4.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

4.1.3 Applicable documents

Document	Revision	Title
SHV-PD-SPEC-0101	Rev A	Component specification
SHV-PD-DWG-0201	current	General arrangement drawing
SHV-PD-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

4.1.4 General requirements

ID	Requirement
PD-MBC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PD-MBC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PD-MBC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PD-MBC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PD-MBC-G05	Documentation: Acceptance data packages shall be delivered with every unit.

4.2 Zone substation (PD-SUB)

Detailed requirements: SHV-PD-SPEC-0102 Rev B. Drawing: SHV-PD-DWG-0202. Bill of materials: SHV-PD-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
PD-SUB-R01	Substation rating	2.5 TW	±2.5%	Demonstration
PD-SUB-R02	Breaker count	450 ea	±2%	Test

Verification approach

- PD-SUB-R01: demonstration per the station verification and test plan.
- PD-SUB-R02: test per the station verification and test plan.

4.2.1 Zone allocation and installation

The zone substation is installed in zones Z03, Z08, Z13, Z14, Z15.

Zone	Sequence	Installed by	Inspection
Z03	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z13	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z14	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z15	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

4.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

4.2.3 Applicable documents

Document	Revision	Title
SHV-PD-SPEC-0102	Rev B	Component specification
SHV-PD-DWG-0202	current	General arrangement drawing
SHV-PD-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

4.2.4 General requirements

ID	Requirement
PD-SUB-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PD-SUB-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PD-SUB-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PD-SUB-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PD-SUB-G05	Documentation: Acceptance data packages shall be delivered with every unit.

4.3 Backup generator bank (PD-BKG)

Detailed requirements: SHV-PD-SPEC-0103 Rev B. Drawing: SHV-PD-DWG-0203. Bill of materials: SHV-PD-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
PD-BKG-R01	Bank output	1.35 TW	±2%	Inspection
PD-BKG-R02	Start time	14 s	±2.5%	Analysis

Verification approach

- PD-BKG-R01: inspection per the station verification and test plan.
- PD-BKG-R02: analysis per the station verification and test plan.

4.3.1 Zone allocation and installation

The backup generator bank is installed in zones Z02, Z03, Z12, Z24.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z03	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z12	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z24	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

4.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

4.3.3 Applicable documents

Document	Revision	Title
SHV-PD-SPEC-0103	Rev B	Component specification
SHV-PD-DWG-0203	current	General arrangement drawing
SHV-PD-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

4.3.4 General requirements

ID	Requirement
PD-BKG-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PD-BKG-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PD-BKG-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PD-BKG-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PD-BKG-G05	Documentation: Acceptance data packages shall be delivered with every unit.

4.4 Storage battery string (PD-SBS)

Detailed requirements: SHV-PD-SPEC-0104 Rev A. Drawing: SHV-PD-DWG-0204. Bill of materials: SHV-PD-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
PD-SBS-R01	Storage capacity	101 GWh	±2.5%	Inspection
PD-SBS-R02	Discharge time	5.0 h	±2.5%	Test

Verification approach

- PD-SBS-R01: inspection per the station verification and test plan.
- PD-SBS-R02: test per the station verification and test plan.

4.4.1 Zone allocation and installation

The storage battery string is installed in zones Z06, Z08, Z18, Z20.

Zone	Sequence	Installed by	Inspection
Z06	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z18	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z20	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

4.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

4.4.3 Applicable documents

Document	Revision	Title
SHV-PD-SPEC-0104	Rev A	Component specification
SHV-PD-DWG-0204	current	General arrangement drawing
SHV-PD-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

4.4.4 General requirements

ID	Requirement
PD-SBS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PD-SBS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PD-SBS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PD-SBS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PD-SBS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

5. Solar Collector Array (SA)

The solar collector array supplements the reactor with photovoltaic panels on tracking drives around the station's sunward hemisphere.

Document	Title
SHV-ICD-SA-PD	Interface Control Document: Solar Collector Array / Power Distribution

5.1 Solar collector panel (SA-SCP)

Detailed requirements: SHV-SA-SPEC-0101 Rev B. Drawing: SHV-SA-DWG-0201. Bill of materials: SHV-SA-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
SA-SCP-R01	Panel area	749 m²	±5%	Demonstration
SA-SCP-R02	Conversion efficiency	31.3 %	±5%	Analysis

Verification approach

- SA-SCP-R01: demonstration per the station verification and test plan.
- SA-SCP-R02: analysis per the station verification and test plan.

5.1.1 Zone allocation and installation

The solar collector panel is installed in zones Z10, Z24.

Zone	Sequence	Installed by	Inspection
Z10	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z24	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

5.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

5.1.3 Applicable documents

Document	Revision	Title
SHV-SA-SPEC-0101	Rev B	Component specification
SHV-SA-DWG-0201	current	General arrangement drawing
SHV-SA-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

5.1.4 General requirements

ID	Requirement
SA-SCP-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SA-SCP-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SA-SCP-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SA-SCP-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SA-SCP-G05	Documentation: Acceptance data packages shall be delivered with every unit.

5.2 Array tracking drive (SA-SAT)

Detailed requirements: SHV-SA-SPEC-0102 Rev D. Drawing: SHV-SA-DWG-0202. Bill of materials: SHV-SA-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
SA-SAT-R01	Pointing accuracy	0.42 deg	±2%	Test
SA-SAT-R02	Slew rate	6.6 deg/min	±5%	Inspection

Verification approach

- SA-SAT-R01: test per the station verification and test plan.
- SA-SAT-R02: inspection per the station verification and test plan.

5.2.1 Zone allocation and installation

The array tracking drive is installed in zones Z04, Z08, Z09, Z12.

Zone	Sequence	Installed by	Inspection
Z04	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z09	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z12	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

5.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

5.2.3 Applicable documents

Document	Revision	Title
SHV-SA-SPEC-0102	Rev D	Component specification
SHV-SA-DWG-0202	current	General arrangement drawing
SHV-SA-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

5.2.4 General requirements

ID	Requirement
SA-SAT-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SA-SAT-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SA-SAT-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SA-SAT-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SA-SAT-G05	Documentation: Acceptance data packages shall be delivered with every unit.

5.3 Power conditioning unit (SA-PCU)

Detailed requirements: SHV-SA-SPEC-0103 Rev B. Drawing: SHV-SA-DWG-0203. Bill of materials: SHV-SA-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
SA-PCU-R01	Rated power	1.8 GW	±2%	Demonstration
SA-PCU-R02	Conversion loss	2.1 %	±2.5%	Test

Verification approach

- SA-PCU-R01: demonstration per the station verification and test plan.
- SA-PCU-R02: test per the station verification and test plan.

5.3.1 Zone allocation and installation

The power conditioning unit is installed in zones Z01, Z07, Z12, Z22.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z07	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z12	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z22	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

5.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

5.3.3 Applicable documents

Document	Revision	Title
SHV-SA-SPEC-0103	Rev B	Component specification
SHV-SA-DWG-0203	current	General arrangement drawing
SHV-SA-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

5.3.4 General requirements

ID	Requirement
SA-PCU-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SA-PCU-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SA-PCU-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SA-PCU-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SA-PCU-G05	Documentation: Acceptance data packages shall be delivered with every unit.

6. Propulsion & Station-Keeping (PR)

Propulsion keeps the station in its assigned orbit and attitude. Station-keeping thrusters and sublight ion engines are commanded by the navigation computer, with attitude control gyroscopes for fine pointing.

Document	Title
SHV-ICD-PD-PR	Interface Control Document: Power Distribution / Propulsion & Station-Keeping

6.1 Station-keeping thruster (PR-SKT)

Detailed requirements: SHV-PR-SPEC-0101 Rev B. Drawing: SHV-PR-DWG-0201. Bill of materials: SHV-PR-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
PR-SKT-R01	Thrust	1,579 kN	±2.5%	Inspection
PR-SKT-R02	Thruster count	84 ea	±2%	Inspection

Verification approach

- PR-SKT-R01: inspection per the station verification and test plan.
- PR-SKT-R02: inspection per the station verification and test plan.

6.1.1 Zone allocation and installation

The station-keeping thruster is installed in zones Z08, Z16, Z19, Z21, Z23.

Zone	Sequence	Installed by	Inspection
Z08	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z16	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z19	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z21	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z23	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

6.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

6.1.3 Applicable documents

Document	Revision	Title
SHV-PR-SPEC-0101	Rev B	Component specification
SHV-PR-DWG-0201	current	General arrangement drawing
SHV-PR-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

6.1.4 General requirements

ID	Requirement
PR-SKT-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PR-SKT-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PR-SKT-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PR-SKT-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PR-SKT-G05	Documentation: Acceptance data packages shall be delivered with every unit.

6.2 Sublight ion engine (PR-SIE)

Detailed requirements: SHV-PR-SPEC-0102 Rev B. Drawing: SHV-PR-DWG-0202. Bill of materials: SHV-PR-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
PR-SIE-R01	Engine count	33 ea	±2%	Analysis
PR-SIE-R02	Thrust per engine	1,065 MN	±2%	Test

Verification approach

- PR-SIE-R01: analysis per the station verification and test plan.
- PR-SIE-R02: test per the station verification and test plan.

6.2.1 Zone allocation and installation

The sublight ion engine is installed in zones Z15, Z20.

Zone	Sequence	Installed by	Inspection
Z15	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z20	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

6.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

6.2.3 Applicable documents

Document	Revision	Title
SHV-PR-SPEC-0102	Rev B	Component specification
SHV-PR-DWG-0202	current	General arrangement drawing
SHV-PR-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

6.2.4 General requirements

ID	Requirement
PR-SIE-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PR-SIE-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PR-SIE-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PR-SIE-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PR-SIE-G05	Documentation: Acceptance data packages shall be delivered with every unit.

6.3 Navigation computer (PR-NAV)

Detailed requirements: SHV-PR-SPEC-0103 Rev B. Drawing: SHV-PR-DWG-0203. Bill of materials: SHV-PR-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
PR-NAV-R01	Orbit solution time	49 s	±2.5%	Inspection
PR-NAV-R02	Ephemeris entries	164 M	±1%	Inspection

Verification approach

- PR-NAV-R01: inspection per the station verification and test plan.
- PR-NAV-R02: inspection per the station verification and test plan.

6.3.1 Zone allocation and installation

The navigation computer is installed in zones Z01, Z04, Z06, Z10, Z11.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z04	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z06	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z10	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z11	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

6.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

6.3.3 Applicable documents

Document	Revision	Title
SHV-PR-SPEC-0103	Rev B	Component specification
SHV-PR-DWG-0203	current	General arrangement drawing
SHV-PR-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

6.3.4 General requirements

ID	Requirement
PR-NAV-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PR-NAV-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PR-NAV-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PR-NAV-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PR-NAV-G05	Documentation: Acceptance data packages shall be delivered with every unit.

6.4 Attitude control gyroscope (PR-ACG)

Detailed requirements: SHV-PR-SPEC-0104 Rev C. Drawing: SHV-PR-DWG-0204. Bill of materials: SHV-PR-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
PR-ACG-R01	Angular momentum	5.1 GN·m·s	±0.5%	Test
PR-ACG-R02	Gyroscope count	8 ea	±5%	Analysis

Verification approach

- PR-ACG-R01: test per the station verification and test plan.
- PR-ACG-R02: analysis per the station verification and test plan.

6.4.1 Zone allocation and installation

The attitude control gyroscope is installed in zones Z02, Z04, Z05, Z10, Z19, Z20.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z04	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z05	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z10	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z19	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z20	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

6.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

6.4.3 Applicable documents

Document	Revision	Title
SHV-PR-SPEC-0104	Rev C	Component specification
SHV-PR-DWG-0204	current	General arrangement drawing
SHV-PR-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

6.4.4 General requirements

ID	Requirement
PR-ACG-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
PR-ACG-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
PR-ACG-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
PR-ACG-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
PR-ACG-G05	Documentation: Acceptance data packages shall be delivered with every unit.

7. Docking & Hangar Bays (HB)

The docking and hangar bays receive, service and launch passenger shuttles, cargo haulers and visiting ships behind magnetic atmosphere containment fields.

Document	Title
SHV-ICD-PD-HB	Interface Control Document: Power Distribution / Docking & Hangar Bays
SHV-ICD-HB-SY	Interface Control Document: Docking & Hangar Bays / Shipyard Slipways
SHV-ICD-HB-CN	Interface Control Document: Docking & Hangar Bays / Computer Core & Data Network
SHV-ICD-SF-HB	Interface Control Document: Superstructure & Frame / Docking & Hangar Bays

7.1 Main hangar bay (HB-MHB)

Detailed requirements: SHV-HB-SPEC-0011 Rev B. Drawing: SHV-HB-DWG-0201. Bill of materials: SHV-HB-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
HB-MHB-R01	Bay count	357 ea	±0.5%	Analysis
HB-MHB-R02	Atmosphere field rating	99.5 %	±2.5%	Test

Verification approach

- HB-MHB-R01: analysis per the station verification and test plan.
- HB-MHB-R02: test per the station verification and test plan.

7.1.1 Zone allocation and installation

The main hangar bay is installed in zones Z04, Z06, Z07, Z08, Z11, Z24.

Zone	Sequence	Installed by	Inspection
Z04	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z06	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z07	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z08	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z11	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z24	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

7.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7.1.3 Applicable documents

Document	Revision	Title
SHV-HB-SPEC-0011	Rev B	Component specification
SHV-HB-DWG-0201	current	General arrangement drawing
SHV-HB-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

7.1.4 General requirements

ID	Requirement
HB-MHB-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HB-MHB-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HB-MHB-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HB-MHB-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HB-MHB-G05	Documentation: Acceptance data packages shall be delivered with every unit.

7.2 Docking bay (HB-DBY)

Detailed requirements: SHV-HB-SPEC-0102 Rev C. Drawing: SHV-HB-DWG-0202. Bill of materials: SHV-HB-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
HB-DBY-R01	Clear span	237 m	±1%	Analysis
HB-DBY-R02	Mooring load rating	1,595 t	±5%	Test

Verification approach

- HB-DBY-R01: analysis per the station verification and test plan.
- HB-DBY-R02: test per the station verification and test plan.

7.2.1 Zone allocation and installation

The docking bay is installed in zones Z12, Z18, Z24.

Zone	Sequence	Installed by	Inspection
Z12	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z18	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z24	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

7.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7.2.3 Applicable documents

Document	Revision	Title
SHV-HB-SPEC-0102	Rev C	Component specification
SHV-HB-DWG-0202	current	General arrangement drawing
SHV-HB-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

7.2.4 General requirements

ID	Requirement
HB-DBY-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HB-DBY-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HB-DBY-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HB-DBY-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HB-DBY-G05	Documentation: Acceptance data packages shall be delivered with every unit.

7.3 Cargo shuttle cradle (HB-CSC)

Detailed requirements: SHV-HB-SPEC-0103 Rev B. Drawing: SHV-HB-DWG-0203. Bill of materials: SHV-HB-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
HB-CSC-R01	Cradle capacity	5 craft	±2%	Demonstration
HB-CSC-R02	Turnaround time	49 min	±1%	Test

Verification approach

- HB-CSC-R01: demonstration per the station verification and test plan.
- HB-CSC-R02: test per the station verification and test plan.

7.3.1 Zone allocation and installation

The cargo shuttle cradle is installed in zones Z04, Z05, Z11, Z18, Z21.

Zone	Sequence	Installed by	Inspection
Z04	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z05	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z11	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z18	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z21	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

7.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7.3.3 Applicable documents

Document	Revision	Title
SHV-HB-SPEC-0103	Rev B	Component specification
SHV-HB-DWG-0203	current	General arrangement drawing
SHV-HB-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

7.3.4 General requirements

ID	Requirement
HB-CSC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HB-CSC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HB-CSC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HB-CSC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HB-CSC-G05	Documentation: Acceptance data packages shall be delivered with every unit.

7.4 Flight control tower (HB-FCT)

Detailed requirements: SHV-HB-SPEC-0104 Rev C. Drawing: SHV-HB-DWG-0204. Bill of materials: SHV-HB-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
HB-FCT-R01	Controller stations	21 ea	±5%	Demonstration

Verification approach

- HB-FCT-R01: demonstration per the station verification and test plan.

7.4.1 Zone allocation and installation

The flight control tower is installed in zones Z02, Z13, Z14, Z19, Z23.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z13	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z14	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z19	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z23	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

7.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7.4.3 Applicable documents

Document	Revision	Title
SHV-HB-SPEC-0104	Rev C	Component specification
SHV-HB-DWG-0204	current	General arrangement drawing
SHV-HB-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

7.4.4 General requirements

ID	Requirement
HB-FCT-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HB-FCT-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HB-FCT-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HB-FCT-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HB-FCT-G05	Documentation: Acceptance data packages shall be delivered with every unit.

8. Shipyard Slipways (SY)

The shipyard builds and refits freighters, liners and survey ships on pressurized slipways and dry docks, served by gantry cranes, a coating shop and outfitting quays.

Document	Title
SHV-ICD-PD-SY	Interface Control Document: Power Distribution / Shipyard Slipways
SHV-ICD-HB-SY	Interface Control Document: Docking & Hangar Bays / Shipyard Slipways
SHV-ICD-SF-SY	Interface Control Document: Superstructure & Frame / Shipyard Slipways

8.1 Slipway (SY-SLW)

Detailed requirements: SHV-SY-SPEC-0101 Rev C. Drawing: SHV-SY-DWG-0201. Bill of materials: SHV-SY-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
SY-SLW-R01	Slipway length	879 m	±2.5%	Analysis
SY-SLW-R02	Maximum hull mass	192,342 t	±1%	Analysis

Verification approach

- SY-SLW-R01: analysis per the station verification and test plan.
- SY-SLW-R02: analysis per the station verification and test plan.

8.1.1 Zone allocation and installation

The slipway is installed in zones Z01, Z10, Z11, Z22.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z10	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z11	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z22	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

8.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

8.1.3 Applicable documents

Document	Revision	Title
SHV-SY-SPEC-0101	Rev C	Component specification
SHV-SY-DWG-0201	current	General arrangement drawing
SHV-SY-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

8.1.4 General requirements

ID	Requirement
SY-SLW-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SY-SLW-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SY-SLW-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SY-SLW-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SY-SLW-G05	Documentation: Acceptance data packages shall be delivered with every unit.

8.2 Gantry crane (SY-GCR)

Detailed requirements: SHV-SY-SPEC-0102 Rev A. Drawing: SHV-SY-DWG-0202. Bill of materials: SHV-SY-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
SY-GCR-R01	Lift capacity	3,800 t	±1%	Analysis
SY-GCR-R02	Crane span	168 m	±2.5%	Inspection

Verification approach

- SY-GCR-R01: analysis per the station verification and test plan.
- SY-GCR-R02: inspection per the station verification and test plan.

8.2.1 Zone allocation and installation

The gantry crane is installed in zones Z01, Z02, Z03, Z06, Z09, Z10.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z02	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z03	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z06	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z09	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z10	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

8.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

8.2.3 Applicable documents

Document	Revision	Title
SHV-SY-SPEC-0102	Rev A	Component specification
SHV-SY-DWG-0202	current	General arrangement drawing
SHV-SY-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

8.2.4 General requirements

ID	Requirement
SY-GCR-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SY-GCR-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SY-GCR-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SY-GCR-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SY-GCR-G05	Documentation: Acceptance data packages shall be delivered with every unit.

8.3 Dry dock (SY-DDK)

Detailed requirements: SHV-SY-SPEC-0103 Rev D. Drawing: SHV-SY-DWG-0203. Bill of materials: SHV-SY-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
SY-DDK-R01	Dock length	1,242 m	±0.5%	Inspection
SY-DDK-R02	Pressurization time	6.6 h	±0.5%	Inspection

Verification approach

- SY-DDK-R01: inspection per the station verification and test plan.
- SY-DDK-R02: inspection per the station verification and test plan.

8.3.1 Zone allocation and installation

The dry dock is installed in zones Z06, Z09.

Zone	Sequence	Installed by	Inspection
Z06	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z09	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

8.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

8.3.3 Applicable documents

Document	Revision	Title
SHV-SY-SPEC-0103	Rev D	Component specification
SHV-SY-DWG-0203	current	General arrangement drawing
SHV-SY-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

8.3.4 General requirements

ID	Requirement
SY-DDK-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SY-DDK-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SY-DDK-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SY-DDK-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SY-DDK-G05	Documentation: Acceptance data packages shall be delivered with every unit.

8.4 Coating shop (SY-PSH)

Detailed requirements: SHV-SY-SPEC-0104 Rev B. Drawing: SHV-SY-DWG-0204. Bill of materials: SHV-SY-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
SY-PSH-R01	Booth count	5 ea	±0.5%	Demonstration
SY-PSH-R02	Cure time	28 h	±2.5%	Analysis

Verification approach

- SY-PSH-R01: demonstration per the station verification and test plan.
- SY-PSH-R02: analysis per the station verification and test plan.

8.4.1 Zone allocation and installation

The coating shop is installed in zones Z04, Z14, Z19.

Zone	Sequence	Installed by	Inspection
Z04	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z14	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z19	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

8.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

8.4.3 Applicable documents

Document	Revision	Title
SHV-SY-SPEC-0104	Rev B	Component specification
SHV-SY-DWG-0204	current	General arrangement drawing
SHV-SY-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

8.4.4 General requirements

ID	Requirement
SY-PSH-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SY-PSH-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SY-PSH-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SY-PSH-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SY-PSH-G05	Documentation: Acceptance data packages shall be delivered with every unit.

8.5 Outfitting quay (SY-OFQ)

Detailed requirements: SHV-SY-SPEC-0105 Rev C. Drawing: SHV-SY-DWG-0205. Bill of materials: SHV-SY-BOM-0305.

Requirement	Parameter	Value	Tolerance	Verification
SY-OFQ-R01	Berths	12 ea	±5%	Test
SY-OFQ-R02	Shore power	217 MW	±1%	Demonstration

Verification approach

- SY-OFQ-R01: test per the station verification and test plan.
- SY-OFQ-R02: demonstration per the station verification and test plan.

8.5.1 Zone allocation and installation

The outfitting quay is installed in zones Z11, Z16, Z17, Z24.

Zone	Sequence	Installed by	Inspection
Z11	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z16	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z17	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z24	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

8.5.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

8.5.3 Applicable documents

Document	Revision	Title
SHV-SY-SPEC-0105	Rev C	Component specification
SHV-SY-DWG-0205	current	General arrangement drawing
SHV-SY-BOM-0305	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

8.5.4 General requirements

ID	Requirement
SY-OFQ-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SY-OFQ-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SY-OFQ-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SY-OFQ-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SY-OFQ-G05	Documentation: Acceptance data packages shall be delivered with every unit.

9. Life Support & Atmospherics (LS)

Life support keeps the atmosphere, water, gravity and temperature within crew limits in every inhabited zone.

Document	Title
SHV-ICD-PD-LS	Interface Control Document: Power Distribution / Life Support & Atmospherics
SHV-ICD-LS-HA	Interface Control Document: Life Support & Atmospherics / Habitation Blocks
SHV-ICD-LS-WM	Interface Control Document: Life Support & Atmospherics / Waste Management & Recycling
SHV-ICD-LS-AG	Interface Control Document: Life Support & Atmospherics / Hydroponics & Food Production

9.1 Atmosphere processor (LS-ATP)

Detailed requirements: SHV-LS-SPEC-0101 Rev C. Drawing: SHV-LS-DWG-0201. Bill of materials: SHV-LS-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
LS-ATP-R01	Processing rate	8,694 m³/s	±2%	Demonstration
LS-ATP-R02	O2 setpoint	21.0 %	±1%	Demonstration

Verification approach

- LS-ATP-R01: demonstration per the station verification and test plan.
- LS-ATP-R02: demonstration per the station verification and test plan.

9.1.1 Zone allocation and installation

The atmosphere processor is installed in zones Z10, Z19, Z23.

Zone	Sequence	Installed by	Inspection
Z10	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z19	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z23	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

9.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

9.1.3 Applicable documents

Document	Revision	Title
SHV-LS-SPEC-0101	Rev C	Component specification
SHV-LS-DWG-0201	current	General arrangement drawing
SHV-LS-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

9.1.4 General requirements

ID	Requirement
LS-ATP-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
LS-ATP-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
LS-ATP-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
LS-ATP-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
LS-ATP-G05	Documentation: Acceptance data packages shall be delivered with every unit.

9.2 Water reclamation plant (LS-WRC)

Detailed requirements: SHV-LS-SPEC-0005 Rev B. Drawing: SHV-LS-DWG-0202. Bill of materials: SHV-LS-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
LS-WRC-R01	Reclamation rate	98.2 %	±5%	Inspection
LS-WRC-R02	Throughput	7,976 m³/h	±2%	Analysis

Verification approach

- LS-WRC-R01: inspection per the station verification and test plan.
- LS-WRC-R02: analysis per the station verification and test plan.

9.2.1 Zone allocation and installation

The water reclamation plant is installed in zones Z19, Z23.

Zone	Sequence	Installed by	Inspection
Z19	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z23	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

9.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

9.2.3 Applicable documents

Document	Revision	Title
SHV-LS-SPEC-0005	Rev B	Component specification
SHV-LS-DWG-0202	current	General arrangement drawing
SHV-LS-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

9.2.4 General requirements

ID	Requirement
LS-WRC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
LS-WRC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
LS-WRC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
LS-WRC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
LS-WRC-G05	Documentation: Acceptance data packages shall be delivered with every unit.

9.3 Artificial gravity generator (LS-GRV)

Detailed requirements: SHV-LS-SPEC-0103 Rev C. Drawing: SHV-LS-DWG-0203. Bill of materials: SHV-LS-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
LS-GRV-R01	Field uniformity	97.8 %	±5%	Inspection

Verification approach

- LS-GRV-R01: inspection per the station verification and test plan.

9.3.1 Zone allocation and installation

The artificial gravity generator is installed in zones Z16, Z17, Z18, Z22, Z24.

Zone	Sequence	Installed by	Inspection
Z16	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z17	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z18	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z22	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z24	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

9.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

9.3.3 Applicable documents

Document	Revision	Title
SHV-LS-SPEC-0103	Rev C	Component specification
SHV-LS-DWG-0203	current	General arrangement drawing
SHV-LS-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

9.3.4 General requirements

ID	Requirement
LS-GRV-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
LS-GRV-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
LS-GRV-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
LS-GRV-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
LS-GRV-G05	Documentation: Acceptance data packages shall be delivered with every unit.

9.4 Thermal regulation loop (LS-TRG)

Detailed requirements: SHV-LS-SPEC-0104 Rev A. Drawing: SHV-LS-DWG-0204. Bill of materials: SHV-LS-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
LS-TRG-R01	Setpoint	21.8 °C	±0.5%	Analysis

Verification approach

- LS-TRG-R01: analysis per the station verification and test plan.

9.4.1 Zone allocation and installation

The thermal regulation loop is installed in zones Z14, Z20.

Zone	Sequence	Installed by	Inspection
Z14	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z20	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

9.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

9.4.3 Applicable documents

Document	Revision	Title
SHV-LS-SPEC-0104	Rev A	Component specification
SHV-LS-DWG-0204	current	General arrangement drawing
SHV-LS-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

9.4.4 General requirements

ID	Requirement
LS-TRG-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
LS-TRG-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
LS-TRG-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
LS-TRG-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
LS-TRG-G05	Documentation: Acceptance data packages shall be delivered with every unit.

10. Habitation Blocks (HA)

The habitation blocks house the resident crew and their families: quarters, mess halls, the medical centre, schools and recreation decks.

Document	Title
SHV-ICD-LS-HA	Interface Control Document: Life Support & Atmospherics / Habitation Blocks
SHV-ICD-HA-AG	Interface Control Document: Habitation Blocks / Hydroponics & Food Production

10.1 Crew quarters module (HA-CRQ)

Detailed requirements: SHV-HA-SPEC-0003 Rev B. Drawing: SHV-HA-DWG-0201. Bill of materials: SHV-HA-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
HA-CRQ-R01	Berths	2,050 ea	±1%	Test

Verification approach

- HA-CRQ-R01: test per the station verification and test plan.

10.1.1 Zone allocation and installation

The crew quarters module is installed in zones Z01, Z06, Z10, Z18, Z21, Z24.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z06	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z10	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z18	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z21	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z24	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

10.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

10.1.3 Applicable documents

Document	Revision	Title
SHV-HA-SPEC-0003	Rev B	Component specification
SHV-HA-DWG-0201	current	General arrangement drawing
SHV-HA-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

10.1.4 General requirements

ID	Requirement
HA-CRQ-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HA-CRQ-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HA-CRQ-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HA-CRQ-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HA-CRQ-G05	Documentation: Acceptance data packages shall be delivered with every unit.

10.2 Mess hall (HA-MSH)

Detailed requirements: SHV-HA-SPEC-0102 Rev D. Drawing: SHV-HA-DWG-0202. Bill of materials: SHV-HA-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
HA-MSH-R01	Seating	871 ea	±5%	Test

Verification approach

- HA-MSH-R01: test per the station verification and test plan.

10.2.1 Zone allocation and installation

The mess hall is installed in zones Z03, Z08, Z11, Z12, Z15.

Zone	Sequence	Installed by	Inspection
Z03	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z11	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z12	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z15	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

10.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

10.2.3 Applicable documents

Document	Revision	Title
SHV-HA-SPEC-0102	Rev D	Component specification
SHV-HA-DWG-0202	current	General arrangement drawing
SHV-HA-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

10.2.4 General requirements

ID	Requirement
HA-MSH-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HA-MSH-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HA-MSH-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HA-MSH-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HA-MSH-G05	Documentation: Acceptance data packages shall be delivered with every unit.

10.3 Medical centre (HA-MED)

Detailed requirements: SHV-HA-SPEC-0103 Rev B. Drawing: SHV-HA-DWG-0203. Bill of materials: SHV-HA-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
HA-MED-R01	Beds	161 ea	±2.5%	Test
HA-MED-R02	Treatment rooms	51 ea	±2%	Analysis

Verification approach

- HA-MED-R01: test per the station verification and test plan.
- HA-MED-R02: analysis per the station verification and test plan.

10.3.1 Zone allocation and installation

The medical centre is installed in zones Z17, Z18.

Zone	Sequence	Installed by	Inspection
Z17	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z18	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

10.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

10.3.3 Applicable documents

Document	Revision	Title
SHV-HA-SPEC-0103	Rev B	Component specification
SHV-HA-DWG-0203	current	General arrangement drawing
SHV-HA-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

10.3.4 General requirements

ID	Requirement
HA-MED-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HA-MED-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HA-MED-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HA-MED-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HA-MED-G05	Documentation: Acceptance data packages shall be delivered with every unit.

10.4 School and childcare wing (HA-SCH)

Detailed requirements: SHV-HA-SPEC-0104 Rev B. Drawing: SHV-HA-DWG-0204. Bill of materials: SHV-HA-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
HA-SCH-R01	Classrooms	31 ea	±5%	Inspection

Verification approach

- HA-SCH-R01: inspection per the station verification and test plan.

10.4.1 Zone allocation and installation

The school and childcare wing is installed in zones Z01, Z03, Z13, Z16, Z20, Z24.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z03	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z13	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z16	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z20	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z24	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

10.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

10.4.3 Applicable documents

Document	Revision	Title
SHV-HA-SPEC-0104	Rev B	Component specification
SHV-HA-DWG-0204	current	General arrangement drawing
SHV-HA-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

10.4.4 General requirements

ID	Requirement
HA-SCH-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HA-SCH-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HA-SCH-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HA-SCH-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HA-SCH-G05	Documentation: Acceptance data packages shall be delivered with every unit.

10.5 Recreation deck (HA-REC)

Detailed requirements: SHV-HA-SPEC-0105 Rev D. Drawing: SHV-HA-DWG-0205. Bill of materials: SHV-HA-BOM-0305.

Requirement	Parameter	Value	Tolerance	Verification
HA-REC-R01	Deck area	15,705 m²	±2%	Demonstration

Verification approach

- HA-REC-R01: demonstration per the station verification and test plan.

10.5.1 Zone allocation and installation

The recreation deck is installed in zones Z01, Z04, Z09, Z20, Z21.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z04	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z09	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z20	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z21	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

10.5.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

10.5.3 Applicable documents

Document	Revision	Title
SHV-HA-SPEC-0105	Rev D	Component specification
SHV-HA-DWG-0205	current	General arrangement drawing
SHV-HA-BOM-0305	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

10.5.4 General requirements

ID	Requirement
HA-REC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
HA-REC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
HA-REC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
HA-REC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
HA-REC-G05	Documentation: Acceptance data packages shall be delivered with every unit.

11. Waste Management & Recycling (WM)

Waste management collects, compacts, treats and recycles station waste through compactor units, treatment plants and a materials recycling line.

Document	Title
SHV-ICD-LS-WM	Interface Control Document: Life Support & Atmospherics / Waste Management & Recycling

11.1 Compactor unit (WM-TCU)

Detailed requirements: SHV-WM-SPEC-0101 Rev B. Drawing: SHV-WM-DWG-0201. Bill of materials: SHV-WM-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
WM-TCU-R01	Compaction force	11.5 MN	±2.5%	Test
WM-TCU-R02	Cycle time	7 min	±2%	Demonstration

Verification approach

- WM-TCU-R01: test per the station verification and test plan.
- WM-TCU-R02: demonstration per the station verification and test plan.

11.1.1 Zone allocation and installation

The compactor unit is installed in zones Z06, Z08, Z12, Z19, Z20.

Zone	Sequence	Installed by	Inspection
Z06	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z12	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z19	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z20	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

11.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

11.1.3 Applicable documents

Document	Revision	Title
SHV-WM-SPEC-0101	Rev B	Component specification
SHV-WM-DWG-0201	current	General arrangement drawing
SHV-WM-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

11.1.4 General requirements

ID	Requirement
WM-TCU-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
WM-TCU-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
WM-TCU-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
WM-TCU-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
WM-TCU-G05	Documentation: Acceptance data packages shall be delivered with every unit.

11.2 Waste treatment plant (WM-WTP)

Detailed requirements: SHV-WM-SPEC-0102 Rev A. Drawing: SHV-WM-DWG-0202. Bill of materials: SHV-WM-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
WM-WTP-R01	Throughput	1,951 t/day	±1%	Test

Verification approach

- WM-WTP-R01: test per the station verification and test plan.

11.2.1 Zone allocation and installation

The waste treatment plant is installed in zones Z04, Z06, Z10.

Zone	Sequence	Installed by	Inspection
Z04	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z06	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z10	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

11.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

11.2.3 Applicable documents

Document	Revision	Title
SHV-WM-SPEC-0102	Rev A	Component specification
SHV-WM-DWG-0202	current	General arrangement drawing
SHV-WM-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

11.2.4 General requirements

ID	Requirement
WM-WTP-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
WM-WTP-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
WM-WTP-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
WM-WTP-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
WM-WTP-G05	Documentation: Acceptance data packages shall be delivered with every unit.

11.3 Materials recycling line (WM-RCY)

Detailed requirements: SHV-WM-SPEC-0103 Rev C. Drawing: SHV-WM-DWG-0203. Bill of materials: SHV-WM-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
WM-RCY-R01	Recovery rate	80.9 %	±5%	Demonstration
WM-RCY-R02	Line throughput	883 t/day	±2.5%	Inspection

Verification approach

- WM-RCY-R01: demonstration per the station verification and test plan.
- WM-RCY-R02: inspection per the station verification and test plan.

11.3.1 Zone allocation and installation

The materials recycling line is installed in zones Z19, Z23.

Zone	Sequence	Installed by	Inspection
Z19	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z23	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

11.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

11.3.3 Applicable documents

Document	Revision	Title
SHV-WM-SPEC-0103	Rev C	Component specification
SHV-WM-DWG-0203	current	General arrangement drawing
SHV-WM-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

11.3.4 General requirements

ID	Requirement
WM-RCY-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
WM-RCY-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
WM-RCY-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
WM-RCY-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
WM-RCY-G05	Documentation: Acceptance data packages shall be delivered with every unit.

12. Hydroponics & Food Production (AG)

Hydroponics and food production grow fresh produce for the resident crew on stacked grow tiers under controlled lighting, and prepare meals in the central kitchens.

Document	Title
SHV-ICD-PD-AG	Interface Control Document: Power Distribution / Hydroponics & Food Production
SHV-ICD-LS-AG	Interface Control Document: Life Support & Atmospherics / Hydroponics & Food Production
SHV-ICD-HA-AG	Interface Control Document: Habitation Blocks / Hydroponics & Food Production

12.1 Hydroponic grow tier (AG-HGT)

Detailed requirements: SHV-AG-SPEC-0101 Rev B. Drawing: SHV-AG-DWG-0201. Bill of materials: SHV-AG-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
AG-HGT-R01	Growing area	6,934 m²	±5%	Test
AG-HGT-R02	Annual yield	103 kg/m²	±1%	Analysis

Verification approach

- AG-HGT-R01: test per the station verification and test plan.
- AG-HGT-R02: analysis per the station verification and test plan.

12.1.1 Zone allocation and installation

The hydroponic grow tier is installed in zones Z08, Z09, Z14.

Zone	Sequence	Installed by	Inspection
Z08	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z09	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z14	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

12.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

12.1.3 Applicable documents

Document	Revision	Title
SHV-AG-SPEC-0101	Rev B	Component specification
SHV-AG-DWG-0201	current	General arrangement drawing
SHV-AG-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

12.1.4 General requirements

ID	Requirement
AG-HGT-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
AG-HGT-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
AG-HGT-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
AG-HGT-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
AG-HGT-G05	Documentation: Acceptance data packages shall be delivered with every unit.

12.2 Nutrient solution dosing unit (AG-NSD)

Detailed requirements: SHV-AG-SPEC-0102 Rev B. Drawing: SHV-AG-DWG-0202. Bill of materials: SHV-AG-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
AG-NSD-R01	Dosing accuracy	2.4 %	±2%	Inspection
AG-NSD-R02	Dosing lines	29 ea	±0.5%	Demonstration

Verification approach

- AG-NSD-R01: inspection per the station verification and test plan.
- AG-NSD-R02: demonstration per the station verification and test plan.

12.2.1 Zone allocation and installation

The nutrient solution dosing unit is installed in zones Z01, Z03, Z04, Z11, Z12, Z21.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z03	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z04	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z11	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z12	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z21	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

12.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

12.2.3 Applicable documents

Document	Revision	Title
SHV-AG-SPEC-0102	Rev B	Component specification
SHV-AG-DWG-0202	current	General arrangement drawing
SHV-AG-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

12.2.4 General requirements

ID	Requirement
AG-NSD-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
AG-NSD-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
AG-NSD-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
AG-NSD-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
AG-NSD-G05	Documentation: Acceptance data packages shall be delivered with every unit.

12.3 Grow lighting array (AG-LGT)

Detailed requirements: SHV-AG-SPEC-0103 Rev B. Drawing: SHV-AG-DWG-0203. Bill of materials: SHV-AG-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
AG-LGT-R01	Photon flux	947 µmol/m²/s	±0.5%	Demonstration
AG-LGT-R02	Fixture count	8,906 ea	±0.5%	Inspection

Verification approach

- AG-LGT-R01: demonstration per the station verification and test plan.
- AG-LGT-R02: inspection per the station verification and test plan.

12.3.1 Zone allocation and installation

The grow lighting array is installed in zones Z03, Z05, Z13, Z14, Z21, Z24.

Zone	Sequence	Installed by	Inspection
Z03	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z05	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z13	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z14	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z21	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z24	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

12.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

12.3.3 Applicable documents

Document	Revision	Title
SHV-AG-SPEC-0103	Rev B	Component specification
SHV-AG-DWG-0203	current	General arrangement drawing
SHV-AG-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

12.3.4 General requirements

ID	Requirement
AG-LGT-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
AG-LGT-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
AG-LGT-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
AG-LGT-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
AG-LGT-G05	Documentation: Acceptance data packages shall be delivered with every unit.

12.4 Food processing kitchen (AG-FPK)

Detailed requirements: SHV-AG-SPEC-0104 Rev C. Drawing: SHV-AG-DWG-0204. Bill of materials: SHV-AG-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
AG-FPK-R01	Meals per day	68,391 ea	±0.5%	Analysis

Verification approach

- AG-FPK-R01: analysis per the station verification and test plan.

12.4.1 Zone allocation and installation

The food processing kitchen is installed in zones Z07, Z09, Z14, Z20.

Zone	Sequence	Installed by	Inspection
Z07	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z09	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z14	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z20	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

12.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

12.4.3 Applicable documents

Document	Revision	Title
SHV-AG-SPEC-0104	Rev C	Component specification
SHV-AG-DWG-0204	current	General arrangement drawing
SHV-AG-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

12.4.4 General requirements

ID	Requirement
AG-FPK-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
AG-FPK-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
AG-FPK-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
AG-FPK-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
AG-FPK-G05	Documentation: Acceptance data packages shall be delivered with every unit.

13. Computer Core & Data Network (CN)

The computer core and data network connect every station system. Maintenance terminals and data ports give technicians and service droids access to the network.

Document	Title
SHV-ICD-HR-CN	Interface Control Document: Hypermatter Reactor / Computer Core & Data Network
SHV-ICD-HB-CN	Interface Control Document: Docking & Hangar Bays / Computer Core & Data Network
SHV-ICD-CN-CM	Interface Control Document: Computer Core & Data Network / Communications Relay
SHV-ICD-CN-OC	Interface Control Document: Computer Core & Data Network / Operations Centre
SHV-ICD-CN-TR	Interface Control Document: Computer Core & Data Network / Internal Transit
SHV-ICD-RL-CN	Interface Control Document: Research Laboratories / Computer Core & Data Network

13.1 Central computer core (CN-CCO)

Detailed requirements: SHV-CN-SPEC-0101 Rev B. Drawing: SHV-CN-DWG-0201. Bill of materials: SHV-CN-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
CN-CCO-R01	Processing nodes	5,485 ea	±0.5%	Test
CN-CCO-R02	Storage	1,899 PB	±0.5%	Demonstration

Verification approach

- CN-CCO-R01: test per the station verification and test plan.
- CN-CCO-R02: demonstration per the station verification and test plan.

13.1.1 Zone allocation and installation

The central computer core is installed in zones Z03, Z05, Z14, Z17, Z22.

Zone	Sequence	Installed by	Inspection
Z03	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z05	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z14	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z17	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z22	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

13.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

13.1.3 Applicable documents

Document	Revision	Title
SHV-CN-SPEC-0101	Rev B	Component specification
SHV-CN-DWG-0201	current	General arrangement drawing
SHV-CN-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

13.1.4 General requirements

ID	Requirement
CN-CCO-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
CN-CCO-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
CN-CCO-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
CN-CCO-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
CN-CCO-G05	Documentation: Acceptance data packages shall be delivered with every unit.

13.2 Data network backbone (CN-DNB)

Detailed requirements: SHV-CN-SPEC-0102 Rev C. Drawing: SHV-CN-DWG-0202. Bill of materials: SHV-CN-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
CN-DNB-R01	Backbone links	234 ea	±2%	Test
CN-DNB-R02	Latency	3.6 ms	±1%	Demonstration

Verification approach

- CN-DNB-R01: test per the station verification and test plan.
- CN-DNB-R02: demonstration per the station verification and test plan.

13.2.1 Zone allocation and installation

The data network backbone is installed in zones Z02, Z03, Z05, Z15, Z23, Z24.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z03	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z05	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z15	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z23	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z24	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

13.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

13.2.3 Applicable documents

Document	Revision	Title
SHV-CN-SPEC-0102	Rev C	Component specification
SHV-CN-DWG-0202	current	General arrangement drawing
SHV-CN-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

13.2.4 General requirements

ID	Requirement
CN-DNB-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
CN-DNB-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
CN-DNB-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
CN-DNB-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
CN-DNB-G05	Documentation: Acceptance data packages shall be delivered with every unit.

13.3 Maintenance data port (CN-MDP)

Detailed requirements: SHV-CN-SPEC-0103 Rev A. Drawing: SHV-CN-DWG-0203. Bill of materials: SHV-CN-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
CN-MDP-R01	Port count	12,761 ea	±0.5%	Inspection

Verification approach

- CN-MDP-R01: inspection per the station verification and test plan.

13.3.1 Zone allocation and installation

The maintenance data port is installed in zones Z04, Z18.

Zone	Sequence	Installed by	Inspection
Z04	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z18	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

13.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

13.3.3 Applicable documents

Document	Revision	Title
SHV-CN-SPEC-0103	Rev A	Component specification
SHV-CN-DWG-0203	current	General arrangement drawing
SHV-CN-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

13.3.4 General requirements

ID	Requirement
CN-MDP-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
CN-MDP-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
CN-MDP-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
CN-MDP-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
CN-MDP-G05	Documentation: Acceptance data packages shall be delivered with every unit.

13.4 Maintenance terminal (CN-MTT)

Detailed requirements: SHV-CN-SPEC-0104 Rev A. Drawing: SHV-CN-DWG-0204. Bill of materials: SHV-CN-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
CN-MTT-R01	Terminal count	4,077 ea	±2.5%	Demonstration

Verification approach

- CN-MTT-R01: demonstration per the station verification and test plan.

13.4.1 Zone allocation and installation

The maintenance terminal is installed in zones Z01, Z06, Z07, Z17, Z19.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z06	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z07	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z17	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z19	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

13.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

13.4.3 Applicable documents

Document	Revision	Title
SHV-CN-SPEC-0104	Rev A	Component specification
SHV-CN-DWG-0204	current	General arrangement drawing
SHV-CN-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

13.4.4 General requirements

ID	Requirement
CN-MTT-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
CN-MTT-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
CN-MTT-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
CN-MTT-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
CN-MTT-G05	Documentation: Acceptance data packages shall be delivered with every unit.

14. Communications Relay (CM)

Communications covers HoloNet transceivers, comm array dishes and the switches that route passenger, cargo and station traffic.

Document	Title
SHV-ICD-CN-CM	Interface Control Document: Computer Core & Data Network / Communications Relay

14.1 HoloNet transceiver (CM-HNT)

Detailed requirements: SHV-CM-SPEC-0101 Rev C. Drawing: SHV-CM-DWG-0201. Bill of materials: SHV-CM-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
CM-HNT-R01	Channels	1,146 ea	±0.5%	Demonstration

Verification approach

- CM-HNT-R01: demonstration per the station verification and test plan.

14.1.1 Zone allocation and installation

The holonet transceiver is installed in zones Z01, Z03, Z13, Z14, Z21.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z03	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z13	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z14	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z21	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

14.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

14.1.3 Applicable documents

Document	Revision	Title
SHV-CM-SPEC-0101	Rev C	Component specification
SHV-CM-DWG-0201	current	General arrangement drawing
SHV-CM-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

14.1.4 General requirements

ID	Requirement
CM-HNT-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
CM-HNT-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
CM-HNT-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
CM-HNT-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
CM-HNT-G05	Documentation: Acceptance data packages shall be delivered with every unit.

14.2 Comm array dish (CM-CMA)

Detailed requirements: SHV-CM-SPEC-0102 Rev C. Drawing: SHV-CM-DWG-0202. Bill of materials: SHV-CM-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
CM-CMA-R01	Dish diameter	112 m	±5%	Analysis
CM-CMA-R02	Array count	12 ea	±2%	Inspection

Verification approach

- CM-CMA-R01: analysis per the station verification and test plan.
- CM-CMA-R02: inspection per the station verification and test plan.

14.2.1 Zone allocation and installation

The comm array dish is installed in zones Z02, Z08, Z12, Z17.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z12	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z17	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

14.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

14.2.3 Applicable documents

Document	Revision	Title
SHV-CM-SPEC-0102	Rev C	Component specification
SHV-CM-DWG-0202	current	General arrangement drawing
SHV-CM-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

14.2.4 General requirements

ID	Requirement
CM-CMA-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
CM-CMA-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
CM-CMA-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
CM-CMA-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
CM-CMA-G05	Documentation: Acceptance data packages shall be delivered with every unit.

14.3 Traffic routing switch (CM-TRS)

Detailed requirements: SHV-CM-SPEC-0103 Rev A. Drawing: SHV-CM-DWG-0203. Bill of materials: SHV-CM-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
CM-TRS-R01	Throughput	8.2 Tbit/s	±1%	Analysis

Verification approach

- CM-TRS-R01: analysis per the station verification and test plan.

14.3.1 Zone allocation and installation

The traffic routing switch is installed in zones Z02, Z04, Z07, Z13, Z23.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z04	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z07	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z13	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z23	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

14.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

14.3.3 Applicable documents

Document	Revision	Title
SHV-CM-SPEC-0103	Rev A	Component specification
SHV-CM-DWG-0203	current	General arrangement drawing
SHV-CM-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

14.3.4 General requirements

ID	Requirement
CM-TRS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
CM-TRS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
CM-TRS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
CM-TRS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
CM-TRS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

15. Superstructure & Frame (SF)

The superstructure is the station's load-bearing frame: structural ribs, hull plating sections, the equatorial service ring and the antenna masts.

Document	Title
SHV-ICD-SF-HB	Interface Control Document: Superstructure & Frame / Docking & Hangar Bays
SHV-ICD-SF-SY	Interface Control Document: Superstructure & Frame / Shipyard Slipways
SHV-ICD-SF-TR	Interface Control Document: Superstructure & Frame / Internal Transit

15.1 Hull plating section (SF-HPS)

Detailed requirements: SHV-SF-SPEC-0101 Rev C. Drawing: SHV-SF-DWG-0201. Bill of materials: SHV-SF-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
SF-HPS-R01	Plate thickness	6.3 m	±5%	Demonstration
SF-HPS-R02	Section count	20,889 ea	±5%	Inspection

Verification approach

- SF-HPS-R01: demonstration per the station verification and test plan.
- SF-HPS-R02: inspection per the station verification and test plan.

15.1.1 Zone allocation and installation

The hull plating section is installed in zones Z07, Z16.

Zone	Sequence	Installed by	Inspection
Z07	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z16	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

15.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

15.1.3 Applicable documents

Document	Revision	Title
SHV-SF-SPEC-0101	Rev C	Component specification
SHV-SF-DWG-0201	current	General arrangement drawing
SHV-SF-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

15.1.4 General requirements

ID	Requirement
SF-HPS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SF-HPS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SF-HPS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SF-HPS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SF-HPS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

15.2 Structural rib (SF-RIB)

Detailed requirements: SHV-SF-SPEC-0102 Rev B. Drawing: SHV-SF-DWG-0202. Bill of materials: SHV-SF-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
SF-RIB-R01	Rib count	612 ea	±2%	Demonstration
SF-RIB-R02	Rib depth	127 m	±1%	Inspection

Verification approach

- SF-RIB-R01: demonstration per the station verification and test plan.
- SF-RIB-R02: inspection per the station verification and test plan.

15.2.1 Zone allocation and installation

The structural rib is installed in zones Z10, Z18, Z21, Z22.

Zone	Sequence	Installed by	Inspection
Z10	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z18	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z21	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z22	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

15.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

15.2.3 Applicable documents

Document	Revision	Title
SHV-SF-SPEC-0102	Rev B	Component specification
SHV-SF-DWG-0202	current	General arrangement drawing
SHV-SF-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

15.2.4 General requirements

ID	Requirement
SF-RIB-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SF-RIB-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SF-RIB-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SF-RIB-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SF-RIB-G05	Documentation: Acceptance data packages shall be delivered with every unit.

15.3 Equatorial service ring (SF-ESR)

Detailed requirements: SHV-SF-SPEC-0103 Rev C. Drawing: SHV-SF-DWG-0203. Bill of materials: SHV-SF-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
SF-ESR-R01	Ring width	790 m	±2%	Test
SF-ESR-R02	Ring depth	307 m	±0.5%	Analysis

Verification approach

- SF-ESR-R01: test per the station verification and test plan.
- SF-ESR-R02: analysis per the station verification and test plan.

15.3.1 Zone allocation and installation

The equatorial service ring is installed in zones Z02, Z04, Z07, Z09, Z12.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z04	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z07	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z09	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z12	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

15.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

15.3.3 Applicable documents

Document	Revision	Title
SHV-SF-SPEC-0103	Rev C	Component specification
SHV-SF-DWG-0203	current	General arrangement drawing
SHV-SF-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

15.3.4 General requirements

ID	Requirement
SF-ESR-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SF-ESR-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SF-ESR-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SF-ESR-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SF-ESR-G05	Documentation: Acceptance data packages shall be delivered with every unit.

15.4 Antenna mast (SF-ANM)

Detailed requirements: SHV-SF-SPEC-0104 Rev B. Drawing: SHV-SF-DWG-0204. Bill of materials: SHV-SF-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
SF-ANM-R01	Mast height	531 m	±1%	Demonstration

Verification approach

- SF-ANM-R01: demonstration per the station verification and test plan.

15.4.1 Zone allocation and installation

The antenna mast is installed in zones Z12, Z13, Z16.

Zone	Sequence	Installed by	Inspection
Z12	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z13	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z16	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

15.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

15.4.3 Applicable documents

Document	Revision	Title
SHV-SF-SPEC-0104	Rev B	Component specification
SHV-SF-DWG-0204	current	General arrangement drawing
SHV-SF-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

15.4.4 General requirements

ID	Requirement
SF-ANM-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
SF-ANM-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
SF-ANM-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
SF-ANM-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
SF-ANM-G05	Documentation: Acceptance data packages shall be delivered with every unit.

16. Internal Transit (TR)

Internal transit moves people and goods around the station: turbolift cars in vertical shafts and a tram that runs around every latitude band.

Document	Title
SHV-ICD-PD-TR	Interface Control Document: Power Distribution / Internal Transit
SHV-ICD-CN-TR	Interface Control Document: Computer Core & Data Network / Internal Transit
SHV-ICD-SF-TR	Interface Control Document: Superstructure & Frame / Internal Transit

16.1 Turbolift car (TR-TLC)

Detailed requirements: SHV-TR-SPEC-0101 Rev C. Drawing: SHV-TR-DWG-0201. Bill of materials: SHV-TR-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
TR-TLC-R01	Car capacity	39 persons	±0.5%	Analysis
TR-TLC-R02	Car speed	11.7 m/s	±2.5%	Demonstration

Verification approach

- TR-TLC-R01: analysis per the station verification and test plan.
- TR-TLC-R02: demonstration per the station verification and test plan.

16.1.1 Zone allocation and installation

The turbolift car is installed in zones Z08, Z14, Z15, Z19, Z21.

Zone	Sequence	Installed by	Inspection
Z08	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z14	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z15	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z19	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z21	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

16.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

16.1.3 Applicable documents

Document	Revision	Title
SHV-TR-SPEC-0101	Rev C	Component specification
SHV-TR-DWG-0201	current	General arrangement drawing
SHV-TR-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

16.1.4 General requirements

ID	Requirement
TR-TLC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TR-TLC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TR-TLC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TR-TLC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TR-TLC-G05	Documentation: Acceptance data packages shall be delivered with every unit.

16.2 Turbolift shaft (TR-TLS)

Detailed requirements: SHV-TR-SPEC-0102 Rev C. Drawing: SHV-TR-DWG-0202. Bill of materials: SHV-TR-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
TR-TLS-R01	Shaft length	8.66 km	±1%	Inspection
TR-TLS-R02	Shaft count	194 ea	±5%	Analysis

Verification approach

- TR-TLS-R01: inspection per the station verification and test plan.
- TR-TLS-R02: analysis per the station verification and test plan.

16.2.1 Zone allocation and installation

The turbolift shaft is installed in zones Z07, Z16, Z22.

Zone	Sequence	Installed by	Inspection
Z07	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z16	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z22	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

16.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

16.2.3 Applicable documents

Document	Revision	Title
SHV-TR-SPEC-0102	Rev C	Component specification
SHV-TR-DWG-0202	current	General arrangement drawing
SHV-TR-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

16.2.4 General requirements

ID	Requirement
TR-TLS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TR-TLS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TR-TLS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TR-TLS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TR-TLS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

16.3 Tram car (TR-TRM)

Detailed requirements: SHV-TR-SPEC-0103 Rev B. Drawing: SHV-TR-DWG-0203. Bill of materials: SHV-TR-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
TR-TRM-R01	Seats	86 ea	±5%	Inspection

Verification approach

- TR-TRM-R01: inspection per the station verification and test plan.

16.3.1 Zone allocation and installation

The tram car is installed in zones Z13, Z19.

Zone	Sequence	Installed by	Inspection
Z13	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z19	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

16.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

16.3.3 Applicable documents

Document	Revision	Title
SHV-TR-SPEC-0103	Rev B	Component specification
SHV-TR-DWG-0203	current	General arrangement drawing
SHV-TR-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

16.3.4 General requirements

ID	Requirement
TR-TRM-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TR-TRM-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TR-TRM-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TR-TRM-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TR-TRM-G05	Documentation: Acceptance data packages shall be delivered with every unit.

16.4 Tram track segment (TR-TRK)

Detailed requirements: SHV-TR-SPEC-0104 Rev D. Drawing: SHV-TR-DWG-0204. Bill of materials: SHV-TR-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
TR-TRK-R01	Segment length	68 m	±5%	Inspection
TR-TRK-R02	Segment count	7,799 ea	±2.5%	Test

Verification approach

- TR-TRK-R01: inspection per the station verification and test plan.
- TR-TRK-R02: test per the station verification and test plan.

16.4.1 Zone allocation and installation

The tram track segment is installed in zones Z05, Z18, Z20, Z21, Z22.

Zone	Sequence	Installed by	Inspection
Z05	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z18	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z20	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z21	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z22	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

16.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

16.4.3 Applicable documents

Document	Revision	Title
SHV-TR-SPEC-0104	Rev D	Component specification
SHV-TR-DWG-0204	current	General arrangement drawing
SHV-TR-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

16.4.4 General requirements

ID	Requirement
TR-TRK-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
TR-TRK-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
TR-TRK-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
TR-TRK-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
TR-TRK-G05	Documentation: Acceptance data packages shall be delivered with every unit.

17. Research Laboratories (RL)

The research laboratories host materials science, biology and astronomy groups: wet and dry labs, cleanroom suites, a zero-gravity laboratory and the station observatory.

Document	Title
SHV-ICD-PD-RL	Interface Control Document: Power Distribution / Research Laboratories
SHV-ICD-RL-CN	Interface Control Document: Research Laboratories / Computer Core & Data Network

17.1 Materials laboratory (RL-MLB)

Detailed requirements: SHV-RL-SPEC-0101 Rev C. Drawing: SHV-RL-DWG-0201. Bill of materials: SHV-RL-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
RL-MLB-R01	Bench count	81 ea	±2.5%	Test

Verification approach

- RL-MLB-R01: test per the station verification and test plan.

17.1.1 Zone allocation and installation

The materials laboratory is installed in zones Z05, Z11, Z14, Z15.

Zone	Sequence	Installed by	Inspection
Z05	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z11	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z14	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z15	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

17.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

17.1.3 Applicable documents

Document	Revision	Title
SHV-RL-SPEC-0101	Rev C	Component specification
SHV-RL-DWG-0201	current	General arrangement drawing
SHV-RL-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

17.1.4 General requirements

ID	Requirement
RL-MLB-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
RL-MLB-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
RL-MLB-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
RL-MLB-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
RL-MLB-G05	Documentation: Acceptance data packages shall be delivered with every unit.

17.2 Cleanroom suite (RL-CLS)

Detailed requirements: SHV-RL-SPEC-0102 Rev D. Drawing: SHV-RL-DWG-0202. Bill of materials: SHV-RL-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
RL-CLS-R01	Cleanroom class	5 class	±1%	Demonstration
RL-CLS-R02	Floor area	1,333 m²	±1%	Demonstration

Verification approach

- RL-CLS-R01: demonstration per the station verification and test plan.
- RL-CLS-R02: demonstration per the station verification and test plan.

17.2.1 Zone allocation and installation

The cleanroom suite is installed in zones Z02, Z05, Z09, Z11, Z18, Z24.

Zone	Sequence	Installed by	Inspection
Z02	Step 1 of 6	Construction Operations	Quality Assurance & Test hold point
Z05	Step 2 of 6	Construction Operations	Quality Assurance & Test hold point
Z09	Step 3 of 6	Construction Operations	Quality Assurance & Test hold point
Z11	Step 4 of 6	Construction Operations	Quality Assurance & Test hold point
Z18	Step 5 of 6	Construction Operations	Quality Assurance & Test hold point
Z24	Step 6 of 6	Construction Operations	Quality Assurance & Test hold point

17.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

17.2.3 Applicable documents

Document	Revision	Title
SHV-RL-SPEC-0102	Rev D	Component specification
SHV-RL-DWG-0202	current	General arrangement drawing
SHV-RL-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

17.2.4 General requirements

ID	Requirement
RL-CLS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
RL-CLS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
RL-CLS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
RL-CLS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
RL-CLS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

17.3 Zero-gravity laboratory (RL-ZGL)

Detailed requirements: SHV-RL-SPEC-0103 Rev A. Drawing: SHV-RL-DWG-0203. Bill of materials: SHV-RL-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
RL-ZGL-R01	Lab volume	5,025 m³	±5%	Inspection

Verification approach

- RL-ZGL-R01: inspection per the station verification and test plan.

17.3.1 Zone allocation and installation

The zero-gravity laboratory is installed in zones Z07, Z09, Z18, Z24.

Zone	Sequence	Installed by	Inspection
Z07	Step 1 of 4	Construction Operations	Quality Assurance & Test hold point
Z09	Step 2 of 4	Construction Operations	Quality Assurance & Test hold point
Z18	Step 3 of 4	Construction Operations	Quality Assurance & Test hold point
Z24	Step 4 of 4	Construction Operations	Quality Assurance & Test hold point

17.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

17.3.3 Applicable documents

Document	Revision	Title
SHV-RL-SPEC-0103	Rev A	Component specification
SHV-RL-DWG-0203	current	General arrangement drawing
SHV-RL-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

17.3.4 General requirements

ID	Requirement
RL-ZGL-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
RL-ZGL-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
RL-ZGL-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
RL-ZGL-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
RL-ZGL-G05	Documentation: Acceptance data packages shall be delivered with every unit.

17.4 Observatory telescope (RL-OBS)

Detailed requirements: SHV-RL-SPEC-0104 Rev A. Drawing: SHV-RL-DWG-0204. Bill of materials: SHV-RL-BOM-0304.

Requirement	Parameter	Value	Tolerance	Verification
RL-OBS-R01	Primary mirror diameter	10.1 m	±0.5%	Analysis
RL-OBS-R02	Pointing accuracy	0.37 arcsec	±5%	Test

Verification approach

- RL-OBS-R01: analysis per the station verification and test plan.
- RL-OBS-R02: test per the station verification and test plan.

17.4.1 Zone allocation and installation

The observatory telescope is installed in zones Z11, Z14.

Zone	Sequence	Installed by	Inspection
Z11	Step 1 of 2	Construction Operations	Quality Assurance & Test hold point
Z14	Step 2 of 2	Construction Operations	Quality Assurance & Test hold point

17.4.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

17.4.3 Applicable documents

Document	Revision	Title
SHV-RL-SPEC-0104	Rev A	Component specification
SHV-RL-DWG-0204	current	General arrangement drawing
SHV-RL-BOM-0304	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

17.4.4 General requirements

ID	Requirement
RL-OBS-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
RL-OBS-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
RL-OBS-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
RL-OBS-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
RL-OBS-G05	Documentation: Acceptance data packages shall be delivered with every unit.

18. Operations Centre (OC)

The operations centre is where station operations, traffic control and facility management are run, with a conference suite and a shared operations display wall.

Document	Title
SHV-ICD-CN-OC	Interface Control Document: Computer Core & Data Network / Operations Centre

18.1 Operations control room (OC-OPC)

Detailed requirements: SHV-OC-SPEC-0101 Rev C. Drawing: SHV-OC-DWG-0201. Bill of materials: SHV-OC-BOM-0301.

Requirement	Parameter	Value	Tolerance	Verification
OC-OPC-R01	Operator stations	41 ea	±2%	Demonstration

Verification approach

- OC-OPC-R01: demonstration per the station verification and test plan.

18.1.1 Zone allocation and installation

The operations control room is installed in zones Z05, Z09, Z15.

Zone	Sequence	Installed by	Inspection
Z05	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z09	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z15	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

18.1.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

18.1.3 Applicable documents

Document	Revision	Title
SHV-OC-SPEC-0101	Rev C	Component specification
SHV-OC-DWG-0201	current	General arrangement drawing
SHV-OC-BOM-0301	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

18.1.4 General requirements

ID	Requirement
OC-OPC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
OC-OPC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
OC-OPC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
OC-OPC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
OC-OPC-G05	Documentation: Acceptance data packages shall be delivered with every unit.

18.2 Conference suite (OC-CRM)

Detailed requirements: SHV-OC-SPEC-0102 Rev C. Drawing: SHV-OC-DWG-0202. Bill of materials: SHV-OC-BOM-0302.

Requirement	Parameter	Value	Tolerance	Verification
OC-CRM-R01	Seats	22 ea	±2.5%	Analysis

Verification approach

- OC-CRM-R01: analysis per the station verification and test plan.

18.2.1 Zone allocation and installation

The conference suite is installed in zones Z01, Z02, Z23.

Zone	Sequence	Installed by	Inspection
Z01	Step 1 of 3	Construction Operations	Quality Assurance & Test hold point
Z02	Step 2 of 3	Construction Operations	Quality Assurance & Test hold point
Z23	Step 3 of 3	Construction Operations	Quality Assurance & Test hold point

18.2.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

18.2.3 Applicable documents

Document	Revision	Title
SHV-OC-SPEC-0102	Rev C	Component specification
SHV-OC-DWG-0202	current	General arrangement drawing
SHV-OC-BOM-0302	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

18.2.4 General requirements

ID	Requirement
OC-CRM-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
OC-CRM-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
OC-CRM-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
OC-CRM-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
OC-CRM-G05	Documentation: Acceptance data packages shall be delivered with every unit.

18.3 Operations display wall (OC-ODW)

Detailed requirements: SHV-OC-SPEC-0103 Rev C. Drawing: SHV-OC-DWG-0203. Bill of materials: SHV-OC-BOM-0303.

Requirement	Parameter	Value	Tolerance	Verification
OC-ODW-R01	Display panels	75 ea	±5%	Demonstration

Verification approach

- OC-ODW-R01: demonstration per the station verification and test plan.

18.3.1 Zone allocation and installation

The operations display wall is installed in zones Z07, Z08, Z11, Z15, Z16.

Zone	Sequence	Installed by	Inspection
Z07	Step 1 of 5	Construction Operations	Quality Assurance & Test hold point
Z08	Step 2 of 5	Construction Operations	Quality Assurance & Test hold point
Z11	Step 3 of 5	Construction Operations	Quality Assurance & Test hold point
Z15	Step 4 of 5	Construction Operations	Quality Assurance & Test hold point
Z16	Step 5 of 5	Construction Operations	Quality Assurance & Test hold point

18.3.2 Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

18.3.3 Applicable documents

Document	Revision	Title
SHV-OC-SPEC-0103	Rev C	Component specification
SHV-OC-DWG-0203	current	General arrangement drawing
SHV-OC-BOM-0303	current	Bill of materials
SHV-QA-PLN-0004	current	Starhaven Verification and Test Plan
SHV-QA-STD-0002	current	Station Workmanship Standard

18.3.4 General requirements

ID	Requirement
OC-ODW-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.
OC-ODW-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.
OC-ODW-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.
OC-ODW-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.
OC-ODW-G05	Documentation: Acceptance data packages shall be delivered with every unit.

Appendix A. Zone allocation matrix, northern hemisphere

Component	Name	Z01	Z02	Z03	Z04	Z05	Z06	Z07	Z08	Z09	Z10	Z11	Z12
HR-RCV	Reactor core vessel					X							
HR-CFG	Containment field generator		X						X				
HR-FIA	Fuel injector assembly										X		
HR-CMF	Coolant manifold	X											
HR-RMS	Reactor monitoring station							X					
TX-HXS	Heat exchanger stack												
TX-RFA	Radiator fin array		X										
TX-CPS	Coolant pump station		X					X					
TX-HPL	Heat pipe loop						X						X
PD-MBC	Main bus conduit		X									X	
PD-SUB	Zone substation			X					X				
PD-BKG	Backup generator bank		X	X									X
PD-SBS	Storage battery string						X		X				
SA-SCP	Solar collector panel										X		
SA-SAT	Array tracking drive				X				X	X			X
SA-PCU	Power conditioning unit	X						X					X
PR-SKT	Station-keeping thruster								X				
PR-SIE	Sublight ion engine												
PR-NAV	Navigation computer	X			X		X				X	X	
PR-ACG	Attitude control gyroscope		X		X	X					X		
HB-MHB	Main hangar bay				X		X	X	X			X	
HB-DBY	Docking bay												X
HB-CSC	Cargo shuttle cradle				X	X						X	
HB-FCT	Flight control tower		X										
SY-SLW	Slipway	X									X	X	
SY-GCR	Gantry crane	X	X	X			X			X	X		
SY-DDK	Dry dock						X			X			
SY-PSH	Coating shop				X								
SY-OFQ	Outfitting quay											X	
LS-ATP	Atmosphere processor										X		

Component	Name	Z01	Z02	Z03	Z04	Z05	Z06	Z07	Z08	Z09	Z10	Z11	Z12
LS-WRC	Water reclamation plant												
LS-GRV	Artificial gravity generator												
LS-TRG	Thermal regulation loop												
HA-CRQ	Crew quarters module	X					X				X		
HA-MSH	Mess hall			X					X			X	X
HA-MED	Medical centre												
HA-SCH	School and childcare wing	X		X									
HA-REC	Recreation deck	X			X					X			
WM-TCU	Compactor unit						X		X				X
WM-WTP	Waste treatment plant				X		X				X		
WM-RCY	Materials recycling line												
AG-HGT	Hydroponic grow tier								X	X			
AG-NSD	Nutrient solution dosing unit	X		X	X							X	X
AG-LGT	Grow lighting array			X		X							
AG-FPK	Food processing kitchen							X		X			
CN-CCO	Central computer core			X		X							
CN-DNB	Data network backbone		X	X		X							
CN-MDP	Maintenance data port				X								
CN-MTT	Maintenance terminal	X					X	X					
CM-HNT	HoloNet transceiver	X		X									
CM-CMA	Comm array dish		X						X				X
CM-TRS	Traffic routing switch		X		X			X					
SF-HPS	Hull plating section							X					
SF-RIB	Structural rib										X		
SF-ESR	Equatorial service ring		X		X			X		X			X
SF-ANM	Antenna mast												X
TR-TLC	Turbolift car								X				
TR-TLS	Turbolift shaft							X					
TR-TRM	Tram car												
TR-TRK	Tram track segment					X							
RL-MLB	Materials laboratory					X						X	

Component	Name	Z01	Z02	Z03	Z04	Z05	Z06	Z07	Z08	Z09	Z10	Z11	Z12
RL-CLS	Cleanroom suite		X			X				X		X	
RL-ZGL	Zero-gravity laboratory							X		X			
RL-OBS	Observatory telescope											X	
OC-OPC	Operations control room					X				X			
OC-CRM	Conference suite	X	X										
OC-ODW	Operations display wall							X	X			X	

Appendix B. Zone allocation matrix, southern hemisphere

Component	Name	Z13	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Z24
HR-RCV	Reactor core vessel												X
HR-CFG	Containment field generator										X		X
HR-FIA	Fuel injector assembly	X									X		
HR-CMF	Coolant manifold							X	X			X	
HR-RMS	Reactor monitoring station									X			
TX-HXS	Heat exchanger stack									X		X	
TX-RFA	Radiator fin array					X							
TX-CPS	Coolant pump station				X	X						X	
TX-HPL	Heat pipe loop		X						X				
PD-MBC	Main bus conduit	X			X				X				
PD-SUB	Zone substation	X	X	X									
PD-BKG	Backup generator bank												X
PD-SBS	Storage battery string						X		X				
SA-SCP	Solar collector panel												X
SA-SAT	Array tracking drive												
SA-PCU	Power conditioning unit										X		
PR-SKT	Station-keeping thruster				X			X		X		X	
PR-SIE	Sublight ion engine			X					X				
PR-NAV	Navigation computer												
PR-ACG	Attitude control gyroscope							X	X				
HB-MHB	Main hangar bay												X
HB-DBY	Docking bay						X						X
HB-CSC	Cargo shuttle cradle						X			X			
HB-FCT	Flight control tower	X	X					X				X	
SY-SLW	Slipway										X		
SY-GCR	Gantry crane												
SY-DDK	Dry dock												
SY-PSH	Coating shop		X					X					
SY-OFQ	Outfitting quay				X	X							X
LS-ATP	Atmosphere processor							X				X	

Component	Name	Z13	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Z24
LS-WRC	Water reclamation plant							X				X	
LS-GRV	Artificial gravity generator				X	X	X				X		X
LS-TRG	Thermal regulation loop		X						X				
HA-CRQ	Crew quarters module						X			X			X
HA-MSH	Mess hall			X									
HA-MED	Medical centre					X	X						
HA-SCH	School and childcare wing	X			X				X				X
HA-REC	Recreation deck								X	X			
WM-TCU	Compactor unit							X	X				
WM-WTP	Waste treatment plant												
WM-RCY	Materials recycling line							X				X	
AG-HGT	Hydroponic grow tier		X										
AG-NSD	Nutrient solution dosing unit									X			
AG-LGT	Grow lighting array	X	X							X			X
AG-FPK	Food processing kitchen		X						X				
CN-CCO	Central computer core		X			X					X		
CN-DNB	Data network backbone			X								X	X
CN-MDP	Maintenance data port						X						
CN-MTT	Maintenance terminal					X		X					
CM-HNT	HoloNet transceiver	X	X							X			
CM-CMA	Comm array dish					X							
CM-TRS	Traffic routing switch	X										X	
SF-HPS	Hull plating section				X								
SF-RIB	Structural rib						X			X	X		
SF-ESR	Equatorial service ring												
SF-ANM	Antenna mast	X			X								
TR-TLC	Turbolift car		X	X				X		X			
TR-TLS	Turbolift shaft				X						X		
TR-TRM	Tram car	X						X					
TR-TRK	Tram track segment						X		X	X	X		
RL-MLB	Materials laboratory		X	X									

Component	Name	Z13	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Z24
RL-CLS	Cleanroom suite						X						X
RL-ZGL	Zero-gravity laboratory						X						X
RL-OBS	Observatory telescope		X										
OC-OPC	Operations control room			X									
OC-CRM	Conference suite											X	
OC-ODW	Operations display wall			X	X								

Appendix C. Requirement index

Requirement	Component	Specification	Value
HR-RCV-R01	Reactor core vessel	SHV-HR-SPEC-0002	1.85 km
HR-RCV-R02	Reactor core vessel	SHV-HR-SPEC-0002	18.7 m
HR-RCV-R03	Reactor core vessel	SHV-HR-SPEC-0002	4,400 K

Requirement	Component	Specification	Value
HR-CFG-R01	Containment field generator	SHV-HR-SPEC-0102	96.66 % nominal
HR-CFG-R02	Containment field generator	SHV-HR-SPEC-0102	19 ea

Requirement	Component	Specification	Value
HR-FIA-R01	Fuel injector assembly	SHV-HR-SPEC-0103	7.79 kg/s
HR-FIA-R02	Fuel injector assembly	SHV-HR-SPEC-0103	128 ea

Requirement	Component	Specification	Value
HR-CMF-R01	Coolant manifold	SHV-HR-SPEC-0104	964 m³/s
HR-CMF-R02	Coolant manifold	SHV-HR-SPEC-0104	42 MPa

Requirement	Component	Specification	Value
HR-RMS-R01	Reactor monitoring station	SHV-HR-SPEC-0105	5,286 ea
HR-RMS-R02	Reactor monitoring station	SHV-HR-SPEC-0105	45 ms

Requirement	Component	Specification	Value
TX-HXS-R01	Heat exchanger stack	SHV-TX-SPEC-0101	0.66 PW
TX-HXS-R02	Heat exchanger stack	SHV-TX-SPEC-0101	20 ea

Requirement	Component	Specification	Value
TX-RFA-R01	Radiator fin array	SHV-TX-SPEC-0007	6.3 km²
TX-RFA-R02	Radiator fin array	SHV-TX-SPEC-0007	0.89

Requirement	Component	Specification	Value
TX-CPS-R01	Coolant pump station	SHV-TX-SPEC-0103	104 m³/s
TX-CPS-R02	Coolant pump station	SHV-TX-SPEC-0103	12 ea

Requirement	Component	Specification	Value
TX-HPL-R01	Heat pipe loop	SHV-TX-SPEC-0104	5.78 km
TX-HPL-R02	Heat pipe loop	SHV-TX-SPEC-0104	282 K

Requirement	Component	Specification	Value
PD-MBC-R01	Main bus conduit	SHV-PD-SPEC-0101	10.3 % reactor
PD-MBC-R02	Main bus conduit	SHV-PD-SPEC-0101	57 km

Requirement	Component	Specification	Value
PD-SUB-R01	Zone substation	SHV-PD-SPEC-0102	2.5 TW
PD-SUB-R02	Zone substation	SHV-PD-SPEC-0102	450 ea

Requirement	Component	Specification	Value
PD-BKG-R01	Backup generator bank	SHV-PD-SPEC-0103	1.35 TW
PD-BKG-R02	Backup generator bank	SHV-PD-SPEC-0103	14 s

Requirement	Component	Specification	Value
PD-SBS-R01	Storage battery string	SHV-PD-SPEC-0104	101 GWh
PD-SBS-R02	Storage battery string	SHV-PD-SPEC-0104	5.0 h

Requirement	Component	Specification	Value
SA-SCP-R01	Solar collector panel	SHV-SA-SPEC-0101	749 m²
SA-SCP-R02	Solar collector panel	SHV-SA-SPEC-0101	31.3 %

Requirement	Component	Specification	Value
SA-SAT-R01	Array tracking drive	SHV-SA-SPEC-0102	0.42 deg
SA-SAT-R02	Array tracking drive	SHV-SA-SPEC-0102	6.6 deg/min

Requirement	Component	Specification	Value
SA-PCU-R01	Power conditioning unit	SHV-SA-SPEC-0103	1.8 GW
SA-PCU-R02	Power conditioning unit	SHV-SA-SPEC-0103	2.1 %

Requirement	Component	Specification	Value
PR-SKT-R01	Station-keeping thruster	SHV-PR-SPEC-0101	1,579 kN
PR-SKT-R02	Station-keeping thruster	SHV-PR-SPEC-0101	84 ea

Requirement	Component	Specification	Value
PR-SIE-R01	Sublight ion engine	SHV-PR-SPEC-0102	33 ea
PR-SIE-R02	Sublight ion engine	SHV-PR-SPEC-0102	1,065 MN

Requirement	Component	Specification	Value
PR-NAV-R01	Navigation computer	SHV-PR-SPEC-0103	49 s
PR-NAV-R02	Navigation computer	SHV-PR-SPEC-0103	164 M

Requirement	Component	Specification	Value
PR-ACG-R01	Attitude control gyroscope	SHV-PR-SPEC-0104	5.1 GN·m·s
PR-ACG-R02	Attitude control gyroscope	SHV-PR-SPEC-0104	8 ea

Requirement	Component	Specification	Value
HB-MHB-R01	Main hangar bay	SHV-HB-SPEC-0011	357 ea
HB-MHB-R02	Main hangar bay	SHV-HB-SPEC-0011	99.5 %

Requirement	Component	Specification	Value
HB-DBY-R01	Docking bay	SHV-HB-SPEC-0102	237 m
HB-DBY-R02	Docking bay	SHV-HB-SPEC-0102	1,595 t

Requirement	Component	Specification	Value
HB-CSC-R01	Cargo shuttle cradle	SHV-HB-SPEC-0103	5 craft
HB-CSC-R02	Cargo shuttle cradle	SHV-HB-SPEC-0103	49 min

Requirement	Component	Specification	Value
HB-FCT-R01	Flight control tower	SHV-HB-SPEC-0104	21 ea

Requirement	Component	Specification	Value
SY-SLW-R01	Slipway	SHV-SY-SPEC-0101	879 m
SY-SLW-R02	Slipway	SHV-SY-SPEC-0101	192,342 t

Requirement	Component	Specification	Value
SY-GCR-R01	Gantry crane	SHV-SY-SPEC-0102	3,800 t
SY-GCR-R02	Gantry crane	SHV-SY-SPEC-0102	168 m

Requirement	Component	Specification	Value
SY-DDK-R01	Dry dock	SHV-SY-SPEC-0103	1,242 m
SY-DDK-R02	Dry dock	SHV-SY-SPEC-0103	6.6 h

Requirement	Component	Specification	Value
SY-PSH-R01	Coating shop	SHV-SY-SPEC-0104	5 ea

Requirement	Component	Specification	Value
SY-PSH-R02	Coating shop	SHV-SY-SPEC-0104	28 h

Requirement	Component	Specification	Value
SY-OFQ-R01	Outfitting quay	SHV-SY-SPEC-0105	12 ea
SY-OFQ-R02	Outfitting quay	SHV-SY-SPEC-0105	217 MW

Requirement	Component	Specification	Value
LS-ATP-R01	Atmosphere processor	SHV-LS-SPEC-0101	8,694 m³/s
LS-ATP-R02	Atmosphere processor	SHV-LS-SPEC-0101	21.0 %

Requirement	Component	Specification	Value
LS-WRC-R01	Water reclamation plant	SHV-LS-SPEC-0005	98.2 %
LS-WRC-R02	Water reclamation plant	SHV-LS-SPEC-0005	7,976 m³/h

Requirement	Component	Specification	Value
LS-GRV-R01	Artificial gravity generator	SHV-LS-SPEC-0103	97.8 %

Requirement	Component	Specification	Value
LS-TRG-R01	Thermal regulation loop	SHV-LS-SPEC-0104	21.8 °C

Requirement	Component	Specification	Value
HA-CRQ-R01	Crew quarters module	SHV-HA-SPEC-0003	2,050 ea

Requirement	Component	Specification	Value
HA-MSH-R01	Mess hall	SHV-HA-SPEC-0102	871 ea

Requirement	Component	Specification	Value
HA-MED-R01	Medical centre	SHV-HA-SPEC-0103	161 ea
HA-MED-R02	Medical centre	SHV-HA-SPEC-0103	51 ea

Requirement	Component	Specification	Value
HA-SCH-R01	School and childcare wing	SHV-HA-SPEC-0104	31 ea

Requirement	Component	Specification	Value
HA-REC-R01	Recreation deck	SHV-HA-SPEC-0105	15,705 m²

Requirement	Component	Specification	Value
WM-TCU-R01	Compactor unit	SHV-WM-SPEC-0101	11.5 MN
WM-TCU-R02	Compactor unit	SHV-WM-SPEC-0101	7 min

Requirement	Component	Specification	Value
WM-WTP-R01	Waste treatment plant	SHV-WM-SPEC-0102	1,951 t/day

Requirement	Component	Specification	Value
WM-RCY-R01	Materials recycling line	SHV-WM-SPEC-0103	80.9 %
WM-RCY-R02	Materials recycling line	SHV-WM-SPEC-0103	883 t/day

Requirement	Component	Specification	Value
AG-HGT-R01	Hydroponic grow tier	SHV-AG-SPEC-0101	6,934 m²
AG-HGT-R02	Hydroponic grow tier	SHV-AG-SPEC-0101	103 kg/m²

Requirement	Component	Specification	Value
AG-NSD-R01	Nutrient solution dosing unit	SHV-AG-SPEC-0102	2.4 %
AG-NSD-R02	Nutrient solution dosing unit	SHV-AG-SPEC-0102	29 ea

Requirement	Component	Specification	Value
AG-LGT-R01	Grow lighting array	SHV-AG-SPEC-0103	947 µmol/m²/s
AG-LGT-R02	Grow lighting array	SHV-AG-SPEC-0103	8,906 ea

Requirement	Component	Specification	Value
AG-FPK-R01	Food processing kitchen	SHV-AG-SPEC-0104	68,391 ea

Requirement	Component	Specification	Value
CN-CCO-R01	Central computer core	SHV-CN-SPEC-0101	5,485 ea
CN-CCO-R02	Central computer core	SHV-CN-SPEC-0101	1,899 PB

Requirement	Component	Specification	Value
CN-DNB-R01	Data network backbone	SHV-CN-SPEC-0102	234 ea
CN-DNB-R02	Data network backbone	SHV-CN-SPEC-0102	3.6 ms

Requirement	Component	Specification	Value
CN-MDP-R01	Maintenance data port	SHV-CN-SPEC-0103	12,761 ea

Requirement	Component	Specification	Value
CN-MTT-R01	Maintenance terminal	SHV-CN-SPEC-0104	4,077 ea

Requirement	Component	Specification	Value
CM-HNT-R01	HoloNet transceiver	SHV-CM-SPEC-0101	1,146 ea

Requirement	Component	Specification	Value
CM-CMA-R01	Comm array dish	SHV-CM-SPEC-0102	112 m
CM-CMA-R02	Comm array dish	SHV-CM-SPEC-0102	12 ea

Requirement	Component	Specification	Value
CM-TRS-R01	Traffic routing switch	SHV-CM-SPEC-0103	8.2 Tbit/s

Requirement	Component	Specification	Value
SF-HPS-R01	Hull plating section	SHV-SF-SPEC-0101	6.3 m
SF-HPS-R02	Hull plating section	SHV-SF-SPEC-0101	20,889 ea

Requirement	Component	Specification	Value
SF-RIB-R01	Structural rib	SHV-SF-SPEC-0102	612 ea
SF-RIB-R02	Structural rib	SHV-SF-SPEC-0102	127 m

Requirement	Component	Specification	Value
SF-ESR-R01	Equatorial service ring	SHV-SF-SPEC-0103	790 m
SF-ESR-R02	Equatorial service ring	SHV-SF-SPEC-0103	307 m

Requirement	Component	Specification	Value
SF-ANM-R01	Antenna mast	SHV-SF-SPEC-0104	531 m

Requirement	Component	Specification	Value
TR-TLC-R01	Turbolift car	SHV-TR-SPEC-0101	39 persons
TR-TLC-R02	Turbolift car	SHV-TR-SPEC-0101	11.7 m/s

Requirement	Component	Specification	Value
TR-TLS-R01	Turbolift shaft	SHV-TR-SPEC-0102	8.66 km
TR-TLS-R02	Turbolift shaft	SHV-TR-SPEC-0102	194 ea

Requirement	Component	Specification	Value
TR-TRM-R01	Tram car	SHV-TR-SPEC-0103	86 ea

Requirement	Component	Specification	Value
TR-TRK-R01	Tram track segment	SHV-TR-SPEC-0104	68 m
TR-TRK-R02	Tram track segment	SHV-TR-SPEC-0104	7,799 ea

Requirement	Component	Specification	Value
RL-MLB-R01	Materials laboratory	SHV-RL-SPEC-0101	81 ea

Requirement	Component	Specification	Value
RL-CLS-R01	Cleanroom suite	SHV-RL-SPEC-0102	5 class
RL-CLS-R02	Cleanroom suite	SHV-RL-SPEC-0102	1,333 m²

Requirement	Component	Specification	Value
RL-ZGL-R01	Zero-gravity laboratory	SHV-RL-SPEC-0103	5,025 m³

Requirement	Component	Specification	Value
RL-OBS-R01	Observatory telescope	SHV-RL-SPEC-0104	10.1 m
RL-OBS-R02	Observatory telescope	SHV-RL-SPEC-0104	0.37 arcsec

Requirement	Component	Specification	Value
OC-OPC-R01	Operations control room	SHV-OC-SPEC-0101	41 ea

Requirement	Component	Specification	Value
OC-CRM-R01	Conference suite	SHV-OC-SPEC-0102	22 ea

Requirement	Component	Specification	Value
OC-ODW-R01	Operations display wall	SHV-OC-SPEC-0103	75 ea